

Support-ticket MCP server

Answers natural-language questions about 40,047 bilingual support tickets — and knows which it cannot answer.

A. DESIGN

- Four MCP tools over stdio, no LLM server-side. Corpus: **40,047** bilingual (en/de) tickets from `Tobi-Bueck/customer-support-tickets` — customer message, first reply, queue label.
- `get_schema` — columns, categorical values, and an explicit `cannot_answer` list (no timestamps, no customer ID, no money) so the host refuses rather than confabulates.
 - `run_sql` — one read-only SELECT against DuckDB. The schema omits the positional `tag_1..8` columns in favour of a long `ticket_tags` table.
 - `find_similar_tickets` — multilingual-e5-small, mean-centred so unrelated pairs score ~0.0 rather than ~0.82; brute-force cosine over **31,625** train-split vectors. Below a **0.47** floor it returns `no_precedent`.
 - `suggest_routing` — k=1 nearest-neighbour vote (k swept on val; larger k floods small queues with the majority class), with two confidence signals: top-1 similarity and top-3 neighbour agreement.

Pipeline: 3 CSVs → 8 stages → DuckDB + npz index, ~80s, deterministic, offline. Every threshold and confidence band is derived on the val split and written to a file the server reads.

B. WHY

The data is free text plus four categorical fields. Retrieval fits the text — *how was something like this handled?* — and SQL fits the fields. Neither answers both kinds of question, so the server does both and the host chooses.

Deliberately avoided:

- An LLM server-side.** The host already has one. A second would double cost per call and make every tool non-deterministic and unvaluable. The one LLM call in the project — labelling 39,739 answers — runs once, offline; its output is committed.
- A fine-tuned classifier for routing.** It would likely score higher; this corpus was published for exactly that. But it returns a softmax with no evidence. kNN returns the neighbours, so a user can check the answer and neighbour agreement becomes a second confidence

D. RISKS

AI agent drafts a reply with no real precedent	Floor 0.47, derived on val: accepts 97.8% of real tickets, rejects 20/20 off-topic queries. Below it, <code>no_precedent</code> and no draft material.
Confidence misread on typed input	Handwritten text scores ~28% lower than the corpus's generated text whatever its topic, so a similarity band calibrated on tickets misreads a typed question. Top-3 agreement does not fall across that shift. Both are returned; when they diverge by >0.15 the response says which to trust.
Source language label wrong on 10% of rows	4,204 tickets labelled <code>de</code> carry English text — one-directional, both source files, found by inspecting one random row. <code>language</code> is re-detected from the text (function words: body, then answer, then label); the original stays as <code>language_label</code> so the defect is queryable. The German score below is on the corrected slice.
Silently wrong tag aggregates	Positional <code>tag_1..8</code> make <code>GROUP BY tag_1</code> run clean and wrong. Those columns are absent from the served schema; <code>ticket_tags</code> holds one row per (ticket, tag), spellings folded, comma-joined cells split.
SQL reads files or mutates	Read-only connection · <code>enable_external_access=false</code> · <code>lock_configuration=true</code> · exactly one SELECT via <code>extract_statements</code> . Capability removed, not function names blacklisted: <code>SELECT * FROM read_csv(...)</code> is a SELECT, and <code>SELECT 1; DROP TABLE x</code> runs both halves in one call.
Memorised test set inflates scores	8,357 tickets appear in two source files. Split first and ~28% of the test set has an identical twin in training. Dedup precedes the split; the index holds train only.
German quietly worse than English	Every metric sliced by language. Found: 0.782 en / 0.542 de macro-F1.

E. BUSINESS IMPACT

The expensive thing in a support queue is elapsed time, not agent minutes. A ticket settled in one exchange costs hours; one that needs three costs days, and the first reply decides which. "*Send the system logs and your account number*" can be acted on. "*Please provide details*" cannot — the customer doesn't know what to send, so an exchange is spent finding out.

49.8% of first replies in this archive are the second kind — they neither resolve the ticket nor name anything fetchable. Each is an avoidable round trip. (*Haiku 4.5 over all 39,739 answers, \$9.53; 55% in a 40-answer hand-labelled sample. Four-way label agreement 70%;*

- signal. And the label space isn't fixed: add a queue, re-index; a classifier retrains.
- A reranker.** The ranking bias that suggested one was a bug: subject-less tickets were 27% of top-1 hits at a 10% base rate, because concatenating `subject + body` made two document populations that don't compete fairly. Embedding body only fixed it for 0.006 macro-F1.

C. PRODUCTION

- Source.** CSVs → DuckDB becomes Zendesk/Jira → Snowflake. The index rebuilds on an Airflow schedule and the abstention floor recalibrates with it — the floor is corpus-relative and coverage grows.
- Transport.** stdio → streamable HTTP behind FastAPI: same tool implementations, second endpoint. `draft_reply` lives there too — a webhook has no host model, so generation moves server-side on that path only.
- Access control.** None today — single-user, local. In production, two layers. SQL is covered by Snowflake row-access policies, *provided the query runs as the caller, not a service account*. The vector index is not: it's a copy of the rows outside the policy boundary, so row policies never see a vector search. Re-gate it separately — resolve the caller's scope to attributes (tenant, region), mask the index, then take top-k. Post-filtering leaks existence and guts top-k for narrow scopes; the `queue filter` already works this way.
- Recency.** A three-year-old precedent may describe a dead product. Recency should weight the ranking, not exclude.
- Scale.** Brute force measures 0.4 ms at 32k vectors, 8 ms at 1M, 41 ms at 5M. Past ~2M, or once the index outgrows one machine's RAM, an ANN index earns its recall cost. Not before — approximate search adds an error inseparable from the embedding's own.
- A limit worth stating.** Nothing here governs whether the host calls a tool, only what it gets back. A model can answer from its own weights, and no floor applies.

the collapsed dead-end / actionable / resolved split 75%, distributionally unbiased — reports/labels.md.)

`reply_state` marks each retrieved precedent, so a draft copies the replies that ended the exchange rather than the ones that extended it. With full threads instead of first replies only, the round-trip count becomes measurable rather than inferred.

At CHEQ. The Data & AI team will be asked to put internal data behind LLM tools. The constraint isn't access — it's whether a stakeholder can trust a number without the data team underwriting it. This repo is the contract for that: refuse what the data can't answer, calibrate every threshold on held-out data, publish the confidence next to the label.